MILITARY SPECIFICATION

ELECTRODES, WELDING, SURFACING, IRON-BASE ALLOY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid wire covered shielded metal arc welding electrodes for depositing iron-base, abrasion resistant weld-metal overlays.

1.2 Classification. Surfacing electrodes shall be furnished in the types, classes and sizes shown in table I, as specified (see 6.2.1).

TABLE 1. Types, classes and sizes.

Type	Class	Positions of welding	Sizes (diameter)
MIL-I-A2a	2	Horizontal and flat	1/8, 5/32
	3	Flat	3/16, 1/4
MIL-1-A2c	2	Horizontal and flat	1/8, 5/32
	3	Flat	3/16, 1/4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-S-22698 -. Steel Plate and Shapes, Weldable Ordinary Strength and Higher Strength: Hull Structural.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized Unit Loads.

2.1.2 Government documents and publications. The following Government document and publication forms a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29,
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-8000 - Surface Inspection Acceptance Standards for Metals.

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

E 10 - Brinell Hardness of Metallic Materials, Test Method for. (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WELDING SOCIETY (AWS)
A3.0 - Welding Terms and Definitions. (DoD adopted)
A5.13 - Solid Surfacing Welding Rods and Electrodes. (DoD adopted)
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
EW 1 - Electric Arc-Welding Apparatus. (DoD adopted)

(Application for copies should be addressed to National Electrical Manufacturers Association, 2101 L Street, NW, Washington; DC 20037.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall produce welding electrodes conforming to the requirements of this specification.

3.1.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Coverings of electrodes.

3.2.1 Handling. Coverings shall withstand ordinary handling without damage that will affect the operation of the electrode.

3,2.2 Concentricity. The coverings on all sizes of electrodes shall be concentric to the extent that maximum core-plus-one covering dimension shall not exceed the minimum core-plus-one covering dimension by more than the following percentages of the mean dimension:

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Electrode size (inch)	Percentage of mean dimension
3/32 and smaller	7
1/8 and larger	5

3.2.3 Dielectric strength. The coverings of MIL-I-A2a electrodes at room temperature and in the dry condition, that is, as removed from freshly opened containers, shall have a dielectric strength sufficient to insulate effectively against a difference potential of 110 volts (V), 60 hertz (Hz), alternating current (a.c.)c The coverings of MIL-I-A2c electrodes are electrically conductive. This electrode type is acceptable under this specification provided the packages are labeled as specified in 5.3.1.1.

3.2.4 Flaking and cracking of covering. The sum of the surface area of the core wire exposed by flaking and cracking of the covering shall not exceed that permitted for arc ends specified in 3.2.8.1, when subjected to the test of 4.4.4.

3.2.5 Slag removal. The slags deposited by the coverings shall be readily removable with hand tools rather than air or power operated tools.

3.2.6 Stability. The stability of the coverings shall be such that electrodes, after receipt from the manufacturer, will comply with this specification after storage in original unopened containers under roof and on dry platforms for the following periods:

Type of unit container	Time period
Class 5 of MIL-W-1043O	Up to 6 months
Classes 1 and 2 of MIL-W-1043O	Up to 1 year

3.2.7 Extent of coverings.

3.2.7.1 Arc ends. The arc end of each electrode shall be sufficiently bare and the covering tapered a minimum amount to permit striking of the electrode. The core wire shall not protrude more than 1/32 inch beyond the covering at the arc end. Chipping of the covering at the arc end may accidentally occur, but when it does occur, such chipping shall not exceed one-quarter the circumference of the core wire and one-quarter of the circumference of the core wire transverse to the arc end.

3.2.7.2 Grip ends. The grip ends shall be bare within the limits specified on figure 1.

3.2.8 Electrode identification.

3.2.8.1 Marking. At least one legible electrode type designation or classification number shall be applied to the electrode covering in such a manner that at least one complete type designation shall come within the space 2-1/2 inches from the grip end of the electrode.

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3.2.8.2 Size and legibility. The imprinted designations shall be composed of equal-size block characters, the height of which shall be approximately 50 to 100 percent of the overall electrode diameter (core wire plus covering) but need not exceed a 5/32 inch height. The imprinted designations shall read from left to right from the grip-end as specified on figure 1. The color of the print shall contrast color of the electrode covering and printed designations shall be readable under normal lighting conditions.

3.2.8.3 Stability. Printed type designations shall remain discernible on electrodes rebaked at temperatures up to 850 degrees Fahrenheit (°F) and on unused portions of partially consumed electrodes or discarded stubs, and shall resist effacement by contact incidental to normal handling, shipping and storing.

3.2.9 Composition. The chemical composition of the covering is optional with the manufacturer. The water content of the covering shall not exceed the percent-by-weight values specified in table II.

TABLE 11. Water content.

Type	Percent by weight maximum
MIL-I-A2a, MIL-I-A2c	0.15

3.2.10 Production equipment. When a change is made in a production run of one MIL-type electrode to another, inspection shall be performed prior to starting the new run to assure that equipment and process lines are purged of material from the previous production run. This inspection shall be performed for all equipment used in manufacturing operations, where the material is not segregated and positively identified, such as cutting to length, extruding, baking, and so forth.

3.3 Core wires of electrodes.

3.3.1 Tolerance in diameter. The diameter of core wire shall not vary more than 0.003 inch from the nominal diameter.

3.3.2 Nominal length. The nominal length of the electrodes shall be in accordance with table 111.

TABLE 111. Nominal length.

Sizes (inch)	Nominal lengths end-grip (inches)
l/8, 5/32	14
3/16, 1/4	14 or 18

3.3.3 Tolerance in length. The actual length of any electrode shall vary not more than l/8-inch from the nominal length specified in table 111.

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3.3.4 Core wire alloy identity. Each end of rod or wire to be spliced during processing shall be tested for alloy identity (see 4.4.9), except when splicing is done to repair a wire break without removing the wire from the process line. Both ends of each coil drawn to finish size shall be tested for alloy identity (see 4.4.9) prior to cutting into electrode core wire lengths.

3.3.5 Covered electrode alloy identity. When specified (see 6.2.1), the core wire of each covered electrode shall be tested for alloy identity after final marking in accordance with 4.4.9.

3.4 Soundness. The weld metal surface shall be free of porosity visible to the unaided eye.

3.5 Linear and non-linear indications. Macro-surfaces of specimens prepared in accordance with 4.4.5 shall meet the acceptance standards for linear and non-linear indications for class 2 welds as specified in NAVSEA 0900-LP-003-8000 . No lack of bond is permitted.

3.6 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition (weight-percent 1/) of deposited weld metal.

Element	MIL-I-A2a	MIL-I-A2c
Carbon	0.30	0.80
Manganese	.50-1.60	1.25-2.50
Phosphorus	.04	0.04
Sulfur	l 03	l 03
Silicon	.20-0.65	.90
Chromium	.65	1.50-3000
Molybdenum	.70	----

1/ Single values are maximum percentages.

3.7 Hardness. The hardness of deposited weld metal shall be 200 to 300 Brinell for MIL-I-A2a and 400 to 600 Brinell for MIL-I-A2c.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specifed herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be as specified in the contract or order (see 6.2.1).

4.2 Quality conformance inspection.

4.2.1 General. Electrodes shall be inspected in accordance with 4.2.1.1 through 4.2.1.6 with lot selected in accordance with 4.2.2.

4.2.1.1 Dry batch. A dry batch of covering mixture is the quantity of dry covering ingredients mixed at one time in one mixing vessel. A dry batch may be used singly or may be subsequently sub-divided into quantities to which the liquid binders may be added to produce a number of wet mixes (see 4.2.1.3).

4.2.1.2 Dry blend. A dry blend is one or more dry batches mixed in a mixing vessel and combined proportionately to produce a uniformity of mixed ingredients equal to that obtained by mixing the same total amount of dry ingredients at one time in one mixing vessel.

4.2.1.3 Wet mix. A wet mix is the combination of a dry batch (see 4.2.1.1) or dry blend (see 4.2.1.2) and liquid binder ingredients at one time in one mixing vessel.

4.2.1.4 Heat of core wire. A heat of core wire is that material obtained from ingots or billets poured from one melt.

4.2.1.5 Controlled core wire chemical composition. Where manufacturing processes are such that core wire is identified by controlled chemical composition for each MIL-type, a lot of electrodes shall be the quantity produced from one or more chemically tested mill coils of rod from one or more heats or metal conforming to the approved chemistry control limits for that type electrode. The chemical composition control limits for mill coils and allowable variation from the standard shall be disclosed by the manufacturer to NAVSEA for evaluation. The following additional conditions shall apply:

(a) Each continuously rolled mill-coiled rod furnished by mills prohibiting spliced-coil practice, shall be chemically analyzed before processing. Mill-coiled rod furnished by mills permitting spliced-coil practice shall have not more than one splice per coil, and both ends of every coil received from such mills shall be chemically analyzed before processing.
(b) The allowable variation (disclosed) from the standard (not disclosed) shall be made available to the Government representative upon request.
(c) Mill coils of rod conforming to established core wire chemistry control for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mixups.

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4.2.1.6 Controlled covering mixture chemical composition. Where manufacturing processes are such that covering mixture is identified by controlled chemical composition for each MIL-type, a lot of electrodes shall be the quantity produced from one or more chemically tested wet mixes conforming to the percent allowable variation from standard for each chemical element for that type electrode. The chemical composition control limits for covering mixture and allowable variation from the standard shall be disclosed by the manufacturer to NAVSEA for evaluation. The following additional conditions shall apply:

(a) Each wet mix shall be chemically analyzed before processing.
(b) The allowable variation (disclosed) from the standard (not disclosed) shall be made available to the Government representative upon request.
(c) Wet mixes conforming to established covering mixture chemistry control for a specific MIL-type electrode shall be appropriately identified and segregated to avoid mixups.

4.2.2 Lot.

4.2.2.1 Lot definition. A lot of electrodes is the quantity of any one size and type produced from one or more wet mixes of covering mixture, as specified in 4.2.1.3 or 4.2.1,6 in conjunction with core wire, and as specified in 4.2.1.4 or 4.2.1.5 during a continuous period of 24 hours or less.

4.2.2.2 Lot identification. Each continuous production period of 24 hours or less shall be identified by the manufacturer's control number or other marking which shall appear on each unit and shipping container. The control number or symbol shall be in a position which permits ready location by the consignee. If the control number or symbol must be decoded for identification, the manufacturer shall furnish the consignee the key or instructions necessary to interpret the code system used.

4.2.3 Sampling and inspection of containers. Unit packages, sealed cans and shipping containers shall be sampled and inspected in accordance with MIL-W-1043O.

4.2.4 Sampling for examination and test of electrodes. Sample electrodes shall be selected either from filled unit packages or from the production line after the baking operation. The unit packages required for this examination may be selected from the sample of packages specified in 4.2.3. If sample electrodes are selected from filled unit packages, the total sample shall be in accordance with MIL-STD-105 for inspection level S-4, with an acceptable quality level (AQL) of 4.0 percent for electrode sizes 1/8 inch and larger and an AQL of 6.5 percent for electrode sizes 3/32 inch and smaller for single sampling. Lot size shall be expressed in pounds. Sample size shall be the number of electrodes to be examined. A minimum number of unit packages shall be selected to permit sampling of ten electrodes maximum from each 50-pound package or from each 10-pound package. If the sample electrodes are selected from the production line, the total sample shall be equivalent to the above and the electrodes shall be selected through the production period so that all parts of the lot are represented. The electrodes selected shall be identified as to type, wet batch, lot, size and other available information such as contract or order number being filled.

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4.2.5 Inspection criteria. The samples selected in accordance with 4.2.4 shall be subjected to the examination of 4.3, the weld metal tests specified in table V, and the tests specified in table VI.

TABLE V. Summary of required weld tests.

Size	Chemical	Hardness	Surface soundness	Macro-examination	Procedures	Requirements
All	x				4.4.7	Table IV
All		x			4.4.5 4.4.6	3*7
All			x		4.4.5	3.4
All				x	4.4.5	3.5

TABLE VI. Summary of other tests.

Tests	Procedures	Requirements
Diameter and length of core wire	4.4.2.1	3.3
Concentricity	4.4.2	3.2.2
Dielectric strength 1/	4.4.3	3.2.3
Covering, flaking and cracking 1/	4.4.4	3.2.4
Covering moisture	4*4.8	3.2.9
Alloy identity - core wire	4.4.9	3*3.4
Alloy identity - finished electrode 1/	4.4.9	3.3.5

1/ When specified (see 6.2.1).

4.3 Visual and dimensional examination. Each of the sample electrodes selected in accordance with 4.2.4 shall be examined to verify conformance with all requirements which do not involve tests specified in 4.4.50

4.4 Test procedure.\

4.4.1 Welding equipment. Welding machines used for supply power for testing electrodes shall be variable voltage direct current (d.c.) motor-generator type, variable voltage d.c. rectifier welder, or a.c. welding transformer conforming to NEMA EW 1. The power source shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 V.

4.4.2 Concentricity measurement. Measurement of the prepared electrode is made with the aid of a supplemental piece of 1/4 by l/4-inch key stock (or equal) 2 + l/16-inch long. The key stock shall be placed symmetrically across the groove and, using any instrument, the core wire diameter plus the thickness of the cover on one side of the electrode plus the thickness of the key stock shall be measured. For instruments using a base for zero reference, the base where it is in contact with the electrode during measurement shall be

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2 + l/16-inch wide and be symmetrical about the groove in the electrode covering. The maximum and minimum dimensions shall be located for each groove of each electrode in the sample measured to the nearest 0.001 inch and recorded. The maximum and minimum pair of dimensions which show the greatest difference for the electrodes required to be measured as specified in 4.2.4 shall be used to determine conformance to 3.2.2.

4.4.2.1 Diameter and length of core wire. In the course of making the measurements specified in 4.4.2, the diameter and lengths of the core wire shall be measured to determine conformance with 3.3.

4.4.3 Dielectric strength. When required (see 6.2.1), dielectric strength of coverings shall be determined by the method specified on figure 2. Methods other than the one shown, yielding the required results, may be used as alternatives if such methods are acceptable to the Government.

4.4.4. Flaking and cracking of covering. When required (see 6.2.1), six electrodes shall be tested for flaking and cracking tendency upon restarting and during welding, in accordance with 4.4.4.1 and 4.4.4.2, respectively. The welding current used shall be in the same range as that used in making the weld test assemblies (see 4.4.5). The welding shall consist of depositing a bead weld on a plate surface.

4.4.4.1 Flaking and cracking upon restart. Three of the six electrodes specified in 4.4.4 shall be tested. The covering shall be grooved to the core wire completely around the electrode at the midpoint of the length. The groove shall be prepared by grinding with the edge of a grind stone (intersection of flat-side of grind stone with its periphery). The core wire shall be uncovered at the root of the groove to form a land 1/16 inch minimum along the length of the electrode. The core wire shall not be undercut. The electrode shall be consumed to the root of the groove in the covering when welding shall be stopped without fusing coating beyond the groove. The half-length electrode shall be removed immediately from the electrode holder and placed upon a flat steel plate to cool until it can be held comfortably in the bare hand. After cooling, the electrode shall be inserted in the electrode holder and welding resumed. After the restart, the half-length electrode shall be consumed for 1 inch of its length. Welding shall then be stopped and the arc end of the electrode examined for conformance to 3.2.4.

4.4.4.2 Flaking and cracking during welding. The remaining three electrodes shall be tested. The electrode shall be consumed to a stub not over 2 inches in length. If flaking or cracking is observed during welding, the welding shall be stopped immediately and the arc end of the electrode examined for conformance to 3,2.4. If no flaking or cracking is observed during welding, the 2-inch stub length of electrode, which is normally discarded, shall not be examined.

4.4.5 Weld test assemblies (weld overlays). Weld test assemblies shall be prepared with sample electrodes obtained in accordance with 4.2.4. Test assemblies shall consist of a minimum overlay of 4 square inches deposited in three layers on a test piece 6 by 4 inches by 3/4 inch thick. The test piece shall conform to MIL-S-22698, grade DH-36 or EH-36T. A record of the welding parameters (welding current, arc voltage, preheat and interpass temperature, and

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number of layers) shall be included on the certificate of test (see 4.6). After cooling, the finished weld surface shall be visually examined for conformance to 3.4, then the surface of the overlay shall be ground smooth and five random hardness readings taken. After hardness testing, the assembly shall be sectioned to provide three coupons for macro-examination at 10X magnification in the polished and etched condition.

4.4.6 Hardness. Hardness shall be measured by the Brinell method specified in ASTM E 10. Rockwell hardness numbers converted to Brinell hardness numbers shall not be used.

4.4.7 Chemical analysis. Chemical analysis shall be made by the testing laboratory or the manufacturer, from a small sample of drilling, milling, chips or slices-machined with tools of high-speed steel from pads of weld metal, for deposits with electrodes deposited on plain low carbon-steel or base material of chemical composition similar to the electrode deposit. The pads of weld metal shall be a minimum of 1-1/2 by 1-1/2 inches by 1/2 inch high for deposits with electrodes up to and including 3/16 inch diameter, and a minimum of 2 by 2 inches by 1/2 inch for deposits with 1/4 inch diameter electrodes. The quantity Of drillings, millings, chips or slices need be no more than necessary for satisfactory analysis. The top surfaces shall be machined off and the samples for chemical analysis shall be machined from the section immediately below, so that no metal. shall be removed within 1/4 inch of the base metal. Chemical analysis shall be made by wet chemical or spectrographic methods. In case of dispute, chemical analysis by wet chemical methods shall be the basis for acceptance. The pad shall be deposited as follows:

(a) Weld metal shall be deposited in multiple pass layers. Each pass shall be deposited, so that its width is 1-1/2 to 2-1/2 times the core wire diameter of the electrode. Welds shall be deposited using common production welding techniques.
(b) The pad may be quenched in hot water (above 180°F) after each pass, to speed cooling. If this technique is employed, the pad should be cooled for approximately 1 minute prior to quenching, and allowed to dry prior to the next pass.
(c) A record of the welding parameter, (welding current, arc voltage, preheat and interpass temperature) base material and approximate number of layers and passes shall be forwarded with the chemical analysis when required by the using activity.

4.4.7.1 Chemical analysis (alternative method). Chemical analysis (wet chemical or spectrographic) may be made on sample drillings, or slices taken from deposited weld metal of groove welds. Extreme care to avoid contamination of the sample shall be exercised when using this method of sampling for chemical analysis; the drillings or slices shall be taken along the centerline of the longitudinal axis of the deposited metal and from the upper third of the groove weld thickness.

4.4.8 Covering moisture. The tendency for the electrode coverings to generate gas, (especially hydrogen) which may be dissolved in the molten weld metal and evolved after solidification of the weld metal, shall be determined by

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analysis of the covering of the middle portion of three electrodes of each size for total water content (hydroscopic and combined) as specified on figure 3 and the notes applying thereto. Electrodes shall be considered unacceptable unless the total water content is in accordance with table II.

4.4.9 Alloy identity. (See 6.6).

4.4.9.1 Procedure. The alloy identity test method may include chemical analysis, metal sorting devices, other approved methods or a combination of methods. The test method shall be submitted for approval to the command or agency concerned.

4.4.9.2 Acceptance criteria. If the test demonstrates that the material is not of the type specified herein, the material shall be rejected.

4.5 Unsatisfactory test results. If the first results of any of the specified tests are determined to be unsatisfactory, two retests for each one that failed shall be permitted. The results of all retests shall be satisfactory for lot acceptance.

4.6 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.2.5, containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot and heat, as applicable, included in a particular shipment, shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The certification shall include test results from the quality conformance test for the shipment. The minimum quality conformance test result data required shall be as listed on figure 4 and may be submitted on the form as shown.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

5.1 Packaging. Packaging of electrodes shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Electrodes shall be packaged in class 1 unit containers in accordance with MIL-W-10430.

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5.1.1.2 Special instructions. When electrodes are required to be packaged in class la unit containers in accordance with MIL-W-1043O, fiberboard or other cushioning material shall be inserted in each end of the unit container to restrict the end-wise movement of the electrodes. The cushioning materials shall be dry and in equilibrium with the electrode covering. The cushioning media may be omitted upon the successful completion of the drop test in accordance with MIL-W-1043O.

5.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and for time periods as specified in 3.2.6.

5.1.2.1 Electrodes shall be packaged in class 1 or 2 containers in accordance with MIL-W-1043O. Unit container weight shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing of electrodes shall be level A, 33 or C, as specified (see 6.2.1).

5.2.1 Levels A and B. Electrodes shall be packed level A or B in accordance with MIL-W-10430, except that for levels A and B packing, palletization for other than cylindrical cans shall be in accordance with MIL-STD-147.

5.2.2 Level C. Packing shall be accomplished in a manner which will ensure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification required by 4.2.2.2, the information shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month and year), and manufacturer's or distributor's brand or type designation. Packages, or the smallest integral unit within a shipping container, shall carry the following warning label, or equivalent, prominently displayed in legible type on the package:

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"DO NOT REMOVE THIS LABEL

WARNING : Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.
Read and understand the manufacturer's instructions and your employer's safety practices
Keep your head out of the fumes.
Use enough general ventilation or exhaust at the arc or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard 249.1 "Safety in Welding and Cutting" published by the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135; OSHA Safety and Health Standards, 29 CFR 1910 are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402."

5.3.1.1 Packages of type MIL-I-A2c electrodes or the smallest integral unit within a shipping container shall carry the following warning label or equivalent, prominently displayed in legible type on the package:

"WARNING: Coverings of MIL-I-A2c electrodes are electrically conductive. Use appropriate safeguards when changing electrodes and during welding."

5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads, in accordance with the manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1, the information shall include shipping destination, stock number as listed by contract, and customer's order number, customer's item number and customer's name.

5.3.3 All containers shall be marked "ASBESTOS FREE".

6. NOTES

6.1 Intended use. Type MIL-I-A2a and MIL-I-A2c welding electrodes are intended for hardfacing applications on carbon and low alloy steels.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Electrodes should be ordered by the pound.
(c) Type, class and size (see 1.2).
(d) whether alloy identity testing of the core wire of each covered electrode after final marking is required (see 3.3.5 and table VI).
(e) Specify inspection system (see 4.1.1).

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(f) Whether dielectric strength test is required (see table VI and 4.4.3).
(g) whether flaking and cracking of covering test is required (see table VI and 4.4.4).
(h) Levels of packaging and packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.6	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired by this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Inspection after delivery. Post delivery inspection of electrodes, to determine conformity to this specification and for acceptance thereof, is the responsibility of the consignee.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Supersession data. Types MIL-FeMn-A and MIL-Fe5-B of MIL-E-19141C dated 16 July 1959 are replaced by EFeMn-A and EFe5-B of AWS A5.13, respectively.

6.6 Mix-up. If, subsequent to shipment, a heat or lot is found to contain material of a type or class different from that specified, the contractor must notify all other customers, who contracted the requirements herein and received material from that heat or lot, of the condition. The contractor is responsible for re-inspection, to ensure that any unused material from the same heat or lot is correct, and must replace all incorrect material at no additional charge.

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6.7 Definitions. For the purpose of this specification, the welding terms and definitions contained in AWS A3.0 apply.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

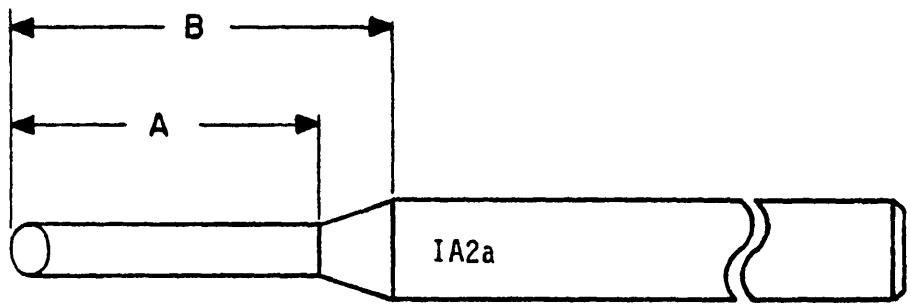
Custodians:
Army - AR
Navy - SH
Air Force - 11

Review activities:
Navy - AS
Air Force - 99
DLA - GS

User activities:
Army - AT
Navy - MC

Preparing activity:
Navy - SH
(Project 3439-0423)

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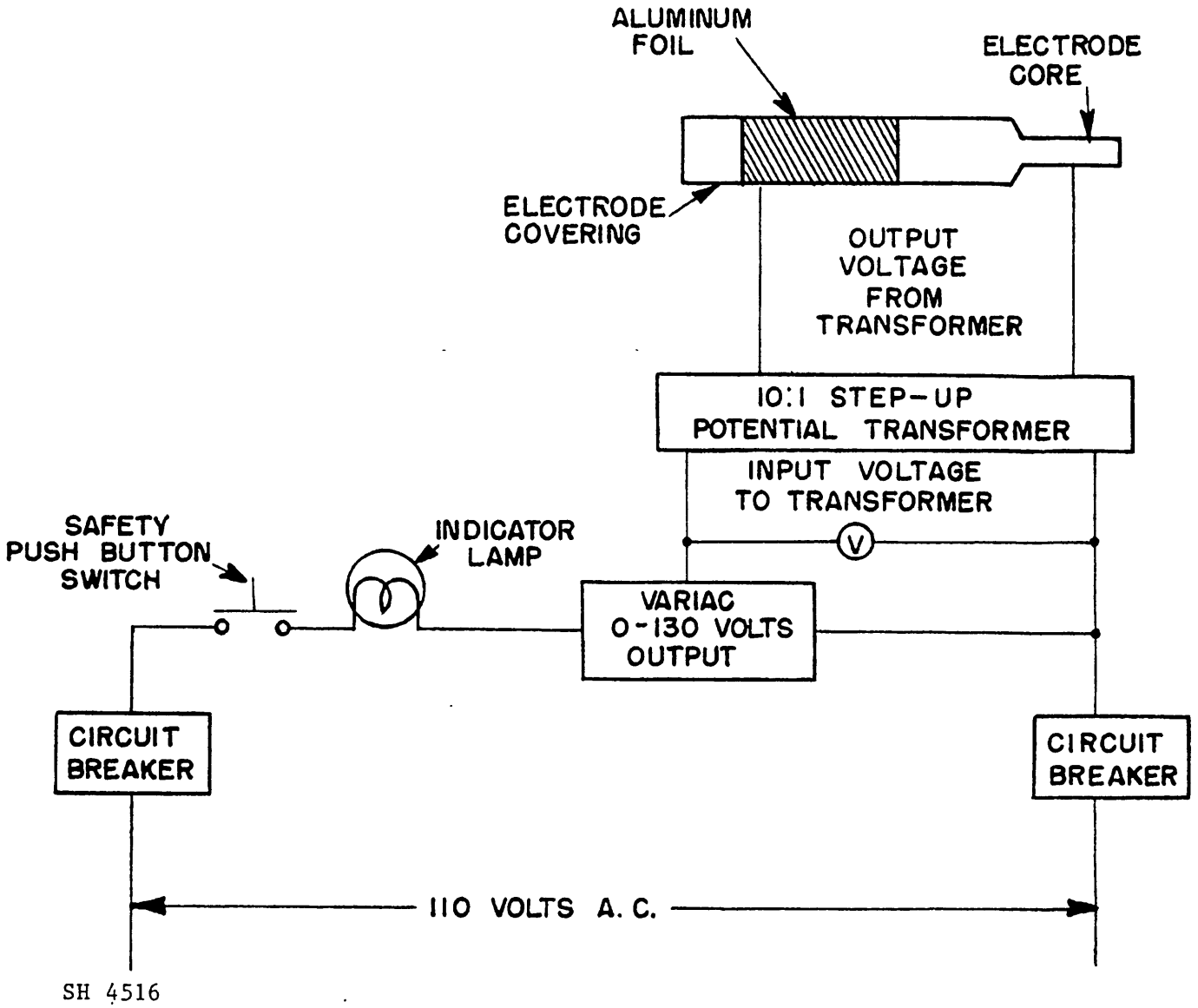


SH 8749

Electrode class (sizes)	Bare portion (minimum) A	Distance to full thickness of covering (maximum) B
	Inch	Inches
1	1/2	1-1/4
2	3/4	1-1/2
3	3/4	1-1/2

FIGURE 1. Grip-end dimensions.

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NOTES:

1. Wrap conducting aluminum foil around 6-inch length of electrode.
2. Place electrode with grip-end on one lug and foil covered section on other.
3. Close circuit breaker; press safety button and slowly rotate variac control.
4. Record maximum voltage before breakdown.
5. Dielectric strength is recorded voltage, multiplied by 10 (transformer ratio).
6. When dielectric strength is below 300 volts, connect appropriate voltmeter across electrode terminals.

FIGURE 2. Circuit for dielectric strength determination.

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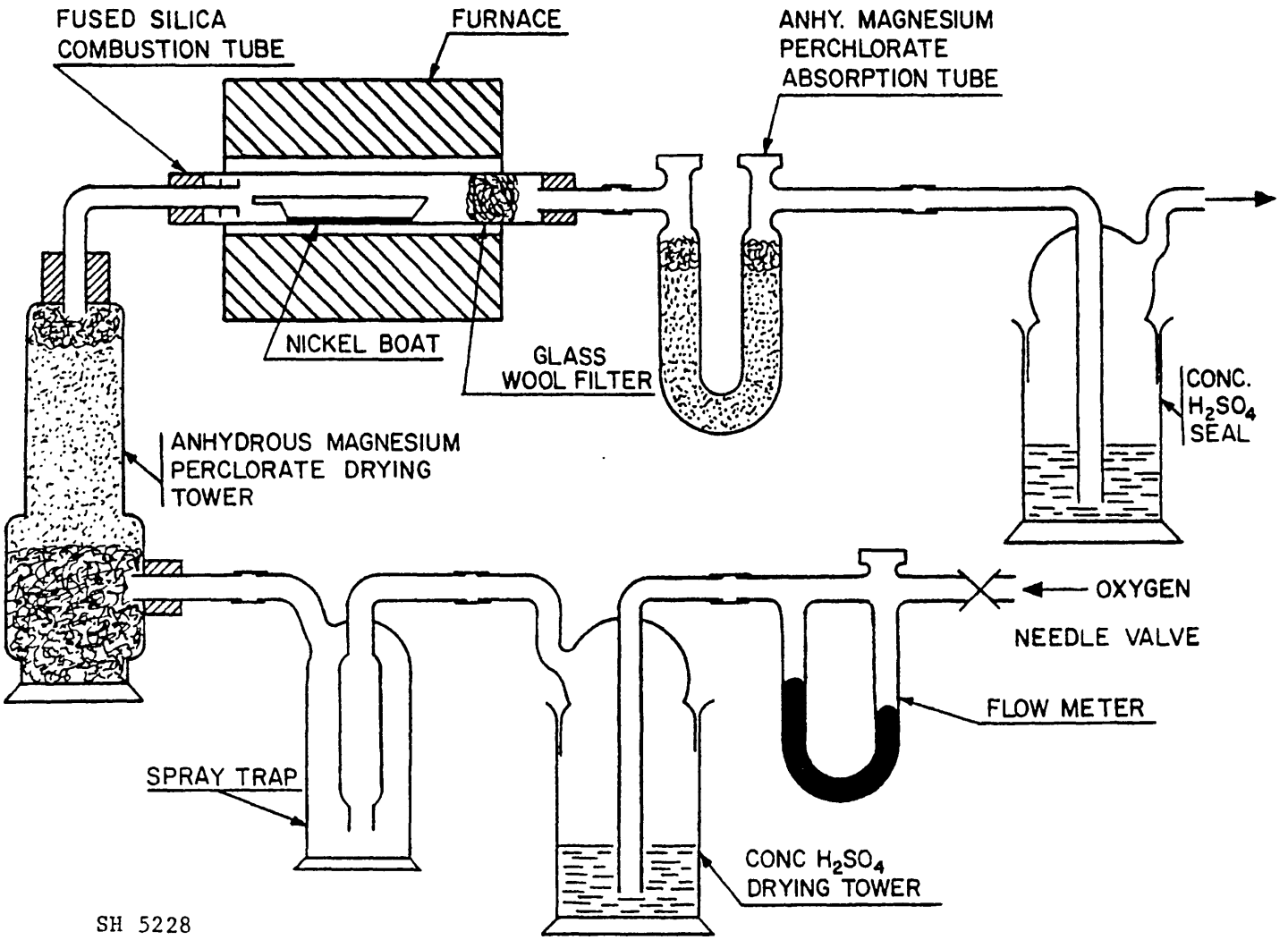


FIGURE 3. Schematic drawing of train for moisture determinations.

NOTES:

1. The method for determining total water content of low-hydrogen type coverings of electric arc welding electrodes shall be as specified below.
2. Apparatus.
   (a) Furnace: Tube furnace capable of furnishing temperatures to 2000°F within the combustion tube is satisfactory. The length of the heating element shall be sufficient to heat 8 to 10 inches of the middle portion of the combustion tube to the required temperature. Furnace shall be equipped with a temperature controller and pyrometer.
   (b) Oxygen supply: The oxygen supply should be free of organic matter. If organic matter is present, oxygen should be purified by passing it through a small combustion tube lightly packed with asbestos or copper oxide, heated to a temperature of 1100 to 1200°F.

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(c) Oxygen drying train: This consists of a pressure reducing valve on the oxygen supply, a needle valve and oxygen flow meter to regulate the flow meter to regulate the flow of oxygen, a wash bottle containing concentrated sulfuric acid (96 percent), a spray trap and a drying tower filled with anhydrous magnesium perchlorate.

(d) Combustion tube: A fused silica combustion tube, open at both ends, with a devitrification point above 2000°F shall be used. About 6 inches of the tube shall project from either side of the furnace. A plug of fine glass wool is inserted into the outlet end of the combustion tube to filter the gases. It shall be inserted far enough into the tube so that-it is heated 400 to 500°F.

(e) Water absorption train: The water driven from the sample is collected in a U-tube absorber (Schwartz) filled with anhydrous magnesium perchlorate. A gas-sealing bottle containing concentrated sulfuric acid (96 percent) is attached to the outlet end of the moisture U-tube absorber.

3. Temperature.

A temperature of 1800°F shall be maintained within the combustion tube for the determination.

4. Preparation and handling of sample.

The sample shall be a composite of the coatings from the middle portions of three electrodes. It should be immediately transferred to a dried, stoppered vial or sample bottle. The sample size used in each determination shall be approximately 4 grams; it shall be weighed directly on the balance dish and transferred to the boat with forceps, weighing the sample to the nearest 0.001 gram.

5. Handling of combustion boat and absorption tube.

The ignited nickel boat after removal from the combustion tube shall be transferred to a Pyrex desiccator containing anhydrous magnesium perchlorate as a desiccant. The absorption tube shall be handled with lint-free gauze at all times and shall be stored in the balance case during the cooling period, so that it assumes the temperature of the atmosphere in which it is to be weighed.

6. Dessicants.

The anhydrous magnesium perchlorate and sulfuric acid shall be renewed often enough to ensure the best performance. In the case of the absorption tube this can be estimated roughly since it is known that anhydrous magnesium perchlorate will absorb at least 16 percent of its weight in water, without any noticeable loss in drying ability.

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7. Oxygen flow.

The flow of oxygen to the train shall be maintained at 200 to 250 milliliters (mL) per minute. Once the flow of oxygen is established, it shall not be changed throughout the determination.

8. Combustion boats.

A nickel combustion boat which will hold a 4-gram sample shall be used. (The ignited sample can be removed after the determination and the boat reused. A small amount of alumina in the heated zone of the combustion tube will prevent nickel boat from fusing with combustion tube.)

9. Step by step procedure.

(a) Blank determination.

(1) With the furnace operating at a temperature of 1800°F, the oxygen flow shall be adjusted to 200 to 250 mL per minute. The nickel boat shall be placed in the middle of the heated zone of the combustion tube and the absorption U-tube attached to the train. A period of 30 minutes shall be employed to ignite the boat and condition the absorption U-tube.

(2) The moisture absorption U-tube shall be removed from the system and placed in the balance case and the nickel boat shall be removed from the combustion tube and placed in the desiccator. The combustion tube of the furnace shall be closed after removing the boat.

(3) After cooling for a period of 20 minutes, the absorption U-tube shall be weighed. The boat shall be removed from the desiccator and exposed for a period of time approximating the time required to transfer a sample from the balance pan to the boat in an actual determination.

(4) The combustion tube shall be opened, the moisture absorption U-tube placed in the system, the boat placed in the center of the combustion tube and the cover replaced.

(5) After a period of 30 minutes, the absorption U-tube shall be removed from the system and placed in the balance case and the boat shall be removed from the combustion tube and placed in the desiccator.

(6) The moisture absorption U-tube shall be weighed after a period of 20 minutes. The gain in weight is the blank.

(b) Moisture determination.

(1) Immediately after weighing the absorption U-tube in item 6 of the blank determination procedure, the sample shall be weighed out on the balance pan.

(2) The boat shall be removed from the desiccator and the sample quickly transferred from the balance pan to the boat.

(3) The combustion tube shall be opened, the moisture absorption U-tube placed in the line, the boat transferred to the center of the combustion tube and the cover replaced.

(4) After a period of 30 minutes, the absorption U-tube shall be removed from the system and placed in the balance case.

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(5) The absorption U-tube shall be weighed after the 20-minute cooling period.

(6) If additional samples are to be run, the nickel boat may be removed from the combustion tube at step 4, the ignited sample is removed and the boat placed in the desiccator to cool. Since the same boat can be used for the following determination, it is not necessary to run a blank determination for each sample, Therefore, after item 5 another determination may be started immediately.

10. Calculation.

$$\text{Percent moisture} = \frac{A-B}{\text{Wt. of sample}} \times 100$$

A = gain in weight of absorption U-tube in moisture determination.
B = gain in weight of absorption U-tube in blank determination.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS 1/

Manufacturer or distributor_____ Customer's name_____

Address_____

Date_____ Customer's order no._____

Specification MIL-_____ Wet batch no._____

Type MIL-_____ Core wire heat no._____

Diameter and length_____

Inspection level_____

Lot no._____

	Welding parameters	Overlay for hardness test	Pads for chemical analysis
Chemical analysis (Complete)	Welding current	_____	_____
Carbon_____	Arc voltage	_____	_____
Manganese_____	Preheat temperature	_____	_____
Phosphorus_____	Interpass temperature	_____	_____
Sulfur_____	Number of layers	_____	_____
Silicon_____	Hardness readings		Macro-examination
Molybdenum_____	1_____ 2_____ 3_____		Satisfactory_____
Chromium_____	4_____ 5_____		

Covering moisture (percent)_____ Concentricity (percent)_____

Covering, iron content (percent)_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized contractor representative

1/ This form is applicable to the quality conformance tests required for each lot. One copy of the quality conformance certificate shall be forwarded for each lot that is included in the shipment.

FIGURE 4. Certification of quality conformance tests.

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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